

Sendov's Conjecture in Degrees Nine, Ten, and Eleven

Principia Math

Abstract

We prove Sendov's conjecture for complex polynomials of degrees nine, ten, and eleven. The three proofs use the same reciprocal normalization. A hypothetical counterexample yields a product constraint, a localization inequality for the mean reciprocal critical-point displacement, and an exact integral identity. After centering the reciprocal critical points, the analytic problem reduces to positivity of explicit bivariate polynomials. Degree nine requires eighteen interior Bernstein certificates and one boundary certificate; degree ten requires fifty interior certificates and one boundary certificate; degree eleven requires a split-integral estimate and 530 interior certificates together with one boundary certificate. Every finite calculation is carried out in exact rational arithmetic, with an independent symbolic reconstruction for each degree.

1 Introduction

Sendov's conjecture asserts that if all zeros of a complex polynomial lie in the closed unit disk, then the closed unit disk centered at each zero contains a zero of the derivative. Brown and Xiang proved the conjecture for degrees at most eight [2]. We establish the next three degrees.

Theorem 1.1. *Let*

$$p(z) = \prod_{k=1}^n (z - z_k), \quad |z_k| \leq 1,$$

where $n \in \{9, 10, 11\}$. All zeros are counted with multiplicity. For every zero a of p , there is a zero ζ of p' such that

$$|a - \zeta| \leq 1.$$

By rotation, the distinguished zero may be taken to be a real number $a \in [0, 1]$. We argue by contradiction and call the polynomial *bad at a* if every zero of p' has distance strictly greater than 1 from a . The proof is computer-assisted only in finite polynomial-positivity checks. The reductions, the certificate polynomials, and the exact verification procedure are all given below.

2 Preliminaries

2.1 Separation of the remaining zeros

We use the Grace–Walsh–Szegő coincidence theorem in its standard form for symmetric multi-affine polynomials on disks; see, for example, [1].

Lemma 2.1. *Let P be a polynomial of degree $n \geq 2$, let α be a zero of P , and suppose that*

$$|\alpha - \zeta| > \rho$$

for every zero ζ of P' . Then α is simple and every other zero β of P satisfies

$$|\beta - \alpha| > 2\rho \sin \frac{\pi}{n}.$$

Proof. The simplicity of α is immediate. Let $\zeta_1, \dots, \zeta_{n-1}$ be the zeros of P' and put $w = \beta - \alpha$. Since $P(\beta) = P(\alpha)$,

$$0 = \int_0^1 \frac{P'(\alpha + tw)}{P'(\alpha)} dt = \int_0^1 \prod_{j=1}^{n-1} (1 - tu_j) dt, \quad u_j = -\frac{w}{\alpha - \zeta_j}.$$

The symmetric multi-affine polynomial

$$F(u_1, \dots, u_{n-1}) = \int_0^1 \prod_{j=1}^{n-1} (1 - tu_j) dt$$

has diagonal

$$F(u, \dots, u) = \frac{1 - (1 - u)^n}{nu},$$

with the value 1 at $u = 0$. Its diagonal zeros are $1 - e^{2\pi i k/n}$, $1 \leq k < n$, and therefore have modulus at least $2 \sin(\pi/n)$. The coincidence theorem implies that F is nonzero when all $|u_j| < 2 \sin(\pi/n)$. If $|w| \leq 2\rho \sin(\pi/n)$, then the strict critical-point separation gives this inequality for every u_j , a contradiction. $\square$

2.2 A reciprocal-square extremum

Lemma 2.2. *Let $0 < m < M$, let $r_1, \dots, r_N \in [m, M]$, and suppose*

$$\prod_{k=1}^N r_k \geq C > m^N.$$

Let ν be the least integer for which $M^\nu m^{N-\nu} \geq C$. Then

$$\sum_{k=1}^N \frac{1}{r_k^2} \leq \frac{N - \nu}{m^2} + \frac{\nu - 1}{M^2} + \left(\frac{M^{\nu-1} m^{N-\nu}}{C} \right)^2.$$

Proof. Put $r_k = e^{t_k}$. Since $t \mapsto e^{-2t}$ is decreasing, we may reduce $\sum t_k$ to $\log C$. On the resulting section of the box $[\log m, \log M]^N$, the convex function $\sum e^{-2t_k}$ is maximized at an extreme point. Hence all but at most one coordinate are endpoints. By the choice of ν , the maximizing configuration has $N - \nu$ entries equal to m , $\nu - 1$ entries equal to M , and one entry equal to $C/(M^{\nu-1} m^{N-\nu})$. $\square$

2.3 Centered elementary symmetric functions

For $d = (d_1, \dots, d_N)$, let $e_m(d)$ denote the m th elementary symmetric function.

Lemma 2.3 (variance under differentiation). *Let f be a monic polynomial of degree $N \geq 2$ whose zeros $d_1, \dots, d_N$ have mean zero, and let $w_1, \dots, w_{N-1}$ be the zeros of f' . Then*

$$\sum_{j=1}^{N-1} |w_j|^2 \leq \frac{N-2}{N} \sum_{j=1}^N |d_j|^2.$$

Proof. Let $A = \text{diag}(d_1, \dots, d_N)$, $u = N^{-1/2}(1, \dots, 1)^\top$, and $P = I - uu^*$. The compression $B = PAP|_{u^\perp}$ has characteristic polynomial f'/N . Indeed, the Schur complement identity gives, first away from the zeros of f and then identically,

$$\det(zI_{u^\perp} - B) = f(z) u^*(zI - A)^{-1}u = \frac{f'(z)}{N}.$$

By Schur triangularization,

$$\sum_{j=1}^{N-1} |w_j|^2 \leq \|B\|_{\text{HS}}^2.$$

Since $u^*Au = 0$ and A is normal,

$$\|B\|_{\text{HS}}^2 = \|PAP\|_{\text{HS}}^2 = \|A\|_{\text{HS}}^2 - \|Au\|^2 - \|A^*u\|^2 = \frac{N-2}{N} \sum_{j=1}^N |d_j|^2.$$

□

Lemma 2.4. *Suppose*

$$\sum_{j=1}^N d_j = 0, \quad \frac{1}{N} \sum_{j=1}^N |d_j|^2 \leq \eta^2.$$

Then

$$|e_2(d)| \leq \frac{N}{2} \eta^2, \tag{1}$$

$$|e_3(d)| \leq \frac{N\sqrt{N-1}}{3} \eta^3, \tag{2}$$

$$|e_m(d)| \leq \left(\frac{N^N}{m^m (N-m)^{N-m}} \right)^{1/2} \eta^m \quad (1 \leq m < N), \tag{3}$$

$$|e_m(d)| \leq \binom{N}{m} \left(\frac{m-1}{N-1} \right)^{m/2} \eta^m \quad (2 \leq m \leq N), \tag{4}$$

$$|e_N(d)| \leq \eta^N. \tag{5}$$

Proof. Newton's identities give $e_2 = -\frac{1}{2} \sum d_j^2$, proving (1). If $V = \sum |d_j|^2$, then

$$\max_j |d_j|^2 \leq \frac{N-1}{N} V,$$

because $d_j = -\sum_{k \neq j} d_k$. Since $e_3 = \frac{1}{3} \sum d_j^3$,

$$|e_3| \leq \frac{1}{3} \max_j |d_j| V \leq \frac{1}{3} \sqrt{\frac{N-1}{N}} V^{3/2},$$

which is (2).

For (3), let $H(z) = \prod_{j=1}^N (1 + zd_j)$. On $|z| = r$, the arithmetic-geometric mean inequality and $\sum d_j = 0$ give

$$|H(z)| \leq \left(\frac{1}{N} \sum_{j=1}^N |1 + zd_j|^2 \right)^{N/2} \leq (1 + r^2 \eta^2)^{N/2}.$$

Cauchy's coefficient estimate yields

$$|e_m(d)| \leq r^{-m}(1 + r^2\eta^2)^{N/2}.$$

For $1 \leq m < N$, minimization at $r^2 = m/((N-m)\eta^2)$ proves (3); the case $\eta = 0$ is immediate.

For (4), let $w_1, \dots, w_m$ be the zeros of $f^{(N-m)}$, where $f(z) = \prod_{j=1}^N (z - d_j)$. Repeated application of Lemma 2.3 gives

$$\frac{1}{m} \sum_{j=1}^m |w_j|^2 \leq \frac{m-1}{N-1} \eta^2.$$

Coefficient comparison gives

$$\prod_{j=1}^m w_j = \frac{e_m(d)}{\binom{N}{m}}.$$

The arithmetic–geometric mean inequality proves (4). Finally, (5) is the arithmetic–geometric mean inequality applied directly to $|d_1|^2, \dots, |d_N|^2$. $\square$

2.4 Bernstein positivity

Lemma 2.5. *Let $R(U, V)$ have bidegree at most (r, s) and write*

$$R(U, V) = \sum_{i=0}^r \sum_{j=0}^s b_{ij} \binom{r}{i} U^i (1-U)^{r-i} \binom{s}{j} V^j (1-V)^{s-j}.$$

If every $b_{ij} > 0$, then $R > 0$ on $[0, 1]^2$.

Proof. The tensor-product Bernstein basis functions are nonnegative and sum to 1. $\square$

3 The reciprocal-integral reduction

Let $n \geq 2$ and put $N = n - 1$. Write

$$p(z) = (z - a) \prod_{k=1}^N (z - z_k), \quad p'(z) = n \prod_{j=1}^N (z - \zeta_j).$$

Assume that p is bad at a . Then a is simple. Set

$$X_k = \frac{1}{a - z_k}, \quad Y_j = \frac{1}{a - \zeta_j}, \quad \mu = \frac{1}{N} \sum_{j=1}^N Y_j = u + iv.$$

Evaluation of p' and p''/p' at a gives

$$\prod_{k=1}^N (a - z_k) = n \prod_{j=1}^N (a - \zeta_j), \quad \sum_{j=1}^N Y_j = 2 \sum_{k=1}^N X_k. \quad (6)$$

Thus

$$\sum_{k=1}^N X_k = \frac{N}{2} \mu, \quad |\mu| < 1. \quad (7)$$

There is also a full coefficient identity. If

$$F(t) = \prod_{k=1}^N (1 - tX_k), \quad G(t) = \prod_{j=1}^N (1 - tY_j),$$

then

$$G(t) = F(t) + tF'(t), \quad e_m(Y) = (m+1)e_m(X) \quad (0 \leq m \leq N). \quad (8)$$

Indeed, $F(t) = q(a-t)/q(a)$ for $p(z) = (z-a)q(z)$, while $G(t) = p'(a-t)/p'(a)$.

Put

$$r_k = |a - z_k|, \quad \sigma = \sum_{k=1}^N r_k^{-2}.$$

Lemma 2.1, the unit-disk hypothesis, and (6) give

$$2 \sin \frac{\pi}{n} < r_k \leq 1 + a, \quad \prod_{k=1}^N r_k > n. \quad (9)$$

Moreover, since $z_k = a - X_k^{-1}$,

$$2a \operatorname{Re} X_k + (1 - a^2)|X_k|^2 = 1 + (1 - |z_k|^2)|X_k|^2 \geq 1.$$

Summing and using (7), we obtain

$$aNu + (1 - a^2)\sigma \geq N. \quad (10)$$

Set

$$\eta^2 = 1 - |\mu|^2, \quad D_j = Y_j - \mu.$$

Then

$$\sum_{j=1}^N D_j = 0, \quad \sum_{j=1}^N |D_j|^2 \leq N\eta^2. \quad (11)$$

Choose constants c_m such that

$$|e_m(D)| \leq c_m \eta^m \quad (2 \leq m \leq N). \quad (12)$$

Finally, define

$$I = \int_0^a G(t) dt = -\frac{p(0)}{p'(a)}. \quad (13)$$

Since $|p(0)| \leq a$ and $|p'(a)| > n$,

$$|I| < \frac{a}{n}. \quad (14)$$

We now state the polynomial criterion used in all three degrees. Suppose that on an interval $[\alpha, \beta]$ we have $\sigma \leq S$. Put

$$L_S(a) = \frac{N - S(1 - a^2)}{aN}, \quad (15)$$

$$q_0(a, \eta; v) = 1 - \frac{2(N - S + Sa^2)}{N}v + a^2(1 - \eta^2)v^2, \quad (16)$$

$$h(a, \eta) = \left(\frac{2S}{N} - 1 \right) (1 - a^2) - a^2\eta^2, \quad (17)$$

$$\ell_m = \left\lfloor \frac{N - m}{2} \right\rfloor, \quad k = \left\lfloor \frac{n}{2} \right\rfloor. \quad (18)$$

For rational S, λ , define

$$P_{n,S,\lambda}(a, \eta) = \frac{1-a}{n} + \frac{\eta^2}{2n} - \frac{h(a, \eta)^k}{n\lambda} - \sum_{m=2}^N c_m a^{m+1} \eta^m \int_0^1 v^m q_0(a, \eta; v)^{\ell_m} dv. \quad (19)$$

This is a polynomial with rational coefficients.

Proposition 3.1 (unsplit certificate). *Assume that, for $a \in [\alpha, \beta]$,*

$$\sigma \leq S, \quad L_S(a) \geq \lambda > 0, \quad Y^2 \geq 1 - \lambda^2, \quad 2\lambda \geq \beta, \quad (20)$$

and

$$0 \leq \frac{2S}{N} - 1, \quad \left(\frac{2S}{N} - 1 \right) (1 - \alpha^2) \leq 1. \quad (21)$$

If

$$P_{n,S,\lambda}(a, \eta) > 0 \quad (\alpha \leq a \leq \beta, \ 0 \leq \eta \leq Y),$$

then no bad polynomial exists with distinguished zero in $[\alpha, \beta]$.

Proof. By (10), $u \geq L_S(a) \geq \lambda$, and hence $0 \leq \eta^2 \leq 1 - \lambda^2 \leq Y^2$. Put

$$J = \int_0^a (1 - t\mu)^N dt = \frac{1 - (1 - a\mu)^n}{n\mu}.$$

For $t = av$,

$$|1 - t\mu|^2 \leq 1 - 2tL_S(a) + (1 - \eta^2)t^2 = q_0(a, \eta; v).$$

The right side is nonnegative, and (20) gives $q_0 \leq 1$. Expanding about μ ,

$$G(t) = (1 - t\mu)^N + \sum_{m=2}^N (-t)^m e_m(D) (1 - t\mu)^{N-m}.$$

Since $0 \leq q_0 \leq 1$, rounding every half-integral exponent down gives

$$|I - J| \leq \sum_{m=2}^N c_m a^{m+1} \eta^m \int_0^1 v^m q_0^{\ell_m} dv. \quad (22)$$

Let $R = |1 - a\mu|$. Then $R^2 \leq h(a, \eta)$. On a feasible pair, $h \geq 0$, while (21) gives $h \leq 1$. Thus $R \leq 1$ and $R^n \leq h^k$. Since $|\mu| \geq \lambda$,

$$|J| \geq \frac{1 - R^n}{n|\mu|} \geq \frac{1}{n} + \frac{\eta^2}{2n} - \frac{h^k}{n\lambda}.$$

Together with (22), this yields

$$|I| - \frac{a}{n} \geq P_{n,S,\lambda}(a, \eta) > 0,$$

contrary to (14). □

For degree eleven we use a split version. Here $N = 10$. Define

$$q_1(a; v) = 1 - \frac{2(N - S + Sa^2)}{N}v + a^2v^2. \quad (23)$$

For $0 \leq \tau \leq 1$, set

$$\begin{aligned} P_{n,S,\lambda}^{(\tau)}(a, \eta) &= \frac{1-a}{n} + \frac{\eta^2}{2n} - \frac{h(a, \eta)^k}{n\lambda} \\ &\quad - \sum_{m=2}^N c_m a^{m+1} \eta^m \int_0^\tau v^m q_0(a, \eta; v)^{\ell_m} dv \\ &\quad - a \int_\tau^1 (q_0(a, \eta; v)^5 + q_1(a; v)^5) dv. \end{aligned} \quad (24)$$

Proposition 3.2 (split certificate). *Let $n = 11$. Under (20) and (21), if*

$$P_{11,S,\lambda}^{(\tau)}(a, \eta) > 0$$

throughout a rectangle contained in $[\alpha, \beta] \times [0, Y]$, then that rectangle contains no feasible pair arising from a bad polynomial.

Proof. On $0 \leq v \leq \tau$, use the termwise estimate from Proposition 3.1. On $\tau \leq v \leq 1$,

$$|G(av)| \leq \left(\frac{1}{N} \sum_{j=1}^N |1 - avY_j|^2 \right)^{N/2} \leq q_1(a; v)^5$$

by the arithmetic–geometric mean inequality, while $|(1 - av\mu)^N| \leq q_0(a, \eta; v)^5$. Thus the last line of (24) bounds the remaining part of $|I - J|$. The estimate for J is unchanged. $\square$

4 Degree nine

Here $N = 8$. We use

$$m_9 = \frac{681}{1000} < 2 \sin \frac{\pi}{9}.$$

Indeed, if $T = 2 \sin(\pi/9)$, then $3T - T^3 = \sqrt{3}$, the function $3t - t^3$ is increasing on $[0, 1]$, and

$$3 - (3m_9 - m_9^3)^2 = \frac{16853534459219919}{10^{18}} > 0.$$

The coefficient majorants are

$$(c_2, \dots, c_8) = \left(4, \frac{264}{35}, 24, 56, 28, 8, 1 \right). \quad (25)$$

The first three follow from Newton’s identities and Lemma 2.4; for $m \geq 5$, Maclaurin’s inequality applied to $|D_1|, \dots, |D_8|$ gives $|e_m(D)| \leq \binom{8}{m} \eta^m$.

4.1 The interval $0 \leq a \leq 9/20$

At $a = 0$, the product inequality in (9) is impossible. For $0 < a \leq 9/20$, apply Lemma 2.2 with $M = 29/20$, $m = m_9$, and $C = 9$. The resulting bound is

$$\sigma < \frac{1101}{200} =: S_0.$$

By (10),

$$u > \frac{8 - S_0(1 - a^2)}{8a}.$$

The right side exceeds 1 on this interval because

$$S_0 a^2 - 8a + 8 - S_0 > 0;$$

the quadratic is decreasing there and has value $781/80000$ at $a = 9/20$. This contradicts $|\mu| < 1$.

4.2 The interval $9/20 \leq a \leq 9/10$

For each row of Table 1, Lemma 2.2, with $M = 1 + \beta$, gives $\sigma \leq S$. The remaining hypotheses of Proposition 3.1 are exact rational inequalities.

Table 1: Degree-nine interior certificate data.

$[\alpha, \beta]$	S	λ	Y
$[9/20, 19/40]$	$11043/2000$	$49/50$	$199/1000$
$[19/40, 1/2]$	$2783/500$	$239/250$	$587/2000$
$[1/2, 21/40]$	$2257/400$	$1863/2000$	$91/250$
$[21/40, 11/20]$	$1151/200$	$1811/2000$	$849/2000$
$[11/20, 23/40]$	$11819/2000$	$879/1000$	$477/1000$
$[23/40, 3/5]$	$764/125$	$1699/2000$	$66/125$
$[3/5, 5/8]$	$637/100$	$817/1000$	$577/1000$
$[5/8, 13/20]$	$13093/2000$	$401/500$	$239/400$
$[13/20, 27/40]$	$13113/2000$	$81/100$	$1173/2000$
$[27/40, 7/10]$	$3289/500$	$409/500$	$1151/2000$
$[7/10, 29/40]$	$529/80$	$413/500$	$141/250$
$[29/40, 3/4]$	$6661/1000$	$1669/2000$	$1103/2000$
$[3/4, 31/40]$	$1681/250$	$843/1000$	$269/500$
$[31/40, 4/5]$	$1701/250$	$213/250$	$131/250$
$[4/5, 33/40]$	$2761/400$	$1723/2000$	$127/250$
$[33/40, 17/20]$	$14041/2000$	$109/125$	$49/100$
$[17/20, 7/8]$	$7161/1000$	$221/250$	$187/400$
$[7/8, 9/10]$	$3663/500$	$359/400$	$883/2000$

After the affine substitution

$$a = \alpha + (\beta - \alpha)U, \quad \eta = YV,$$

the eighteen polynomials (19) have bidegree at most $(9, 8)$. All 1620 exact Bernstein coefficients are positive. The least coefficient is greater than $1/2100$. Proposition 3.1 excludes the entire interval.

4.3 The interval $9/10 \leq a \leq 1$

At $a = 1$, (10) gives $u \geq 1$. Assume $9/10 \leq a < 1$ and put $\delta = 1 - a$. Lemma 2.2, with $M = 2$, gives

$$\sigma < \frac{39}{5}.$$

Consequently

$$1 - u \leq \frac{19}{20}\delta, \quad \eta^2 \leq \frac{19}{10}\delta - \frac{361}{400}\delta^2.$$

Write

$$\delta = x^2, \quad \eta = xy.$$

Then

$$0 < x < \frac{8}{25}, \quad 0 \leq y < \frac{7}{5}.$$

For $0 \leq t \leq 1 - x^2$, put

$$q(x, y; t) = (1 - t)^2 + x^2 t \left(\frac{19}{10} - y^2 t \right).$$

The preceding estimate gives $|1 - t\mu|^2 \leq q \leq 1$. Define

$$\mathcal{E}(x, y) = \sum_{m=2}^8 c_m x^{m-2} y^m \int_0^{1-x^2} t^m q(x, y; t)^{\lfloor (8-m)/2 \rfloor} dt.$$

The centered expansion gives $|I - J| \leq \delta \mathcal{E}$, where $J = \int_0^a (1 - t\mu)^8 dt$. Moreover,

$$|J| \geq \frac{1}{9} + \frac{\eta^2}{18} - \frac{\delta}{1000}.$$

The last estimate follows from $|1 - a\mu|^2 \leq (19/10)\delta$, $|\mu| \geq 181/200$, and the exact integer inequality

$$200^2 19^9 < 1629^2 10^{10}.$$

Thus

$$|I| - \frac{a}{9} \geq \delta \mathcal{P}(x, y), \quad \mathcal{P}(x, y) = \frac{1}{9} + \frac{y^2}{18} - \frac{1}{1000} - \mathcal{E}(x, y).$$

The polynomial $\mathcal{P}$ has bidegree $(24, 8)$. After $x = (8/25)U$, $y = (7/5)V$, all 225 exact Bernstein coefficients exceed $1/100$. This contradicts (14) and completes degree nine.

5 Degree ten

Here $N = 9$. We use

$$m_{10} = \frac{309}{500} < 2 \sin \frac{\pi}{10} = \frac{\sqrt{5} - 1}{2},$$

which is equivalent to $(2m_{10} + 1)^2 < 5$. The coefficient majorants are

m	2	3	4	5	6	7	8	9
c_m	9/2	9	2201/100	2201/100	877/50	217/20	481/100	1.

(26)

The cases $m = 2, 3, 9$ follow from Lemma 2.4; the remaining rational numbers dominate the Cauchy bound (3), as checked by exact squaring.

For $0 \leq a \leq 29/100$,

$$\prod_{k=1}^9 r_k \leq \left(\frac{129}{100}\right)^9 < 10,$$

contrary to (9). On $[29/100, 2/5]$, Lemma 2.2 with $M = 7/5$ has $\nu = 9$ and gives

$$S_0 = 8 \left(\frac{5}{7}\right)^2 + \left(\frac{(7/5)^8}{10}\right)^2.$$

The quadratic

$$S_0 a^2 - 9a + 9 - S_0$$

has positive Bernstein coefficients on $[29/100, 2/5]$. Hence (10) gives $u > 1$.

It remains to cover $[2/5, 9/10]$. For $i = 40, \dots, 89$, set

$$\alpha_i = \frac{i}{100}, \quad \beta_i = \frac{i+1}{100},$$

and let S_i be the bound from Lemma 2.2 with $M = 1 + \beta_i$, $m = m_{10}$, and $C = 10$. Define λ_i to be the largest multiple of 10^{-5} for which

$$S_i a^2 - 9\lambda_i a + 9 - S_i \geq 0 \quad (\alpha_i \leq a \leq \beta_i),$$

certified by its degree-two Bernstein coefficients. Let Y_i be the least multiple of 10^{-5} satisfying $Y_i^2 \geq 1 - \lambda_i^2$.

For each i , all hypotheses of Proposition 3.1 hold, and the exact Bernstein coefficients of

$$P_{10, S_i, \lambda_i}(a, \eta)$$

are positive on $[\alpha_i, \beta_i] \times [0, Y_i]$. These polynomials have bidegree at most $(10, 10)$. Across all fifty rectangles, the least coefficient is greater than $1/3100$.

For the boundary interval, take the bound S_b from Lemma 2.2 with $M = 2$ and set

$$\lambda_b = \frac{22}{25}.$$

The hypotheses of Proposition 3.1 hold on $[9/10, 1]$. Moreover, localization gives

$$1 - u \leq C_b(1 - a), \quad C_b = \frac{S_b(1 + 9/10) - 9}{9(9/10)},$$

and hence $\eta^2 \leq 2C_b(1 - a)$. Put $1 - a = x^2$ and $\eta = xY$. The exact inequalities

$$\left(\frac{8}{25}\right)^2 \geq \frac{1}{10}, \quad \left(\frac{31}{20}\right)^2 \geq 2C_b$$

show that every feasible pair lies in

$$0 \leq x \leq \frac{8}{25}, \quad 0 \leq Y \leq \frac{31}{20}.$$

Divide $P_{10, S_b, \lambda_b}(1 - x^2, xY)$ by x^2 . The quotient is a rational polynomial, and all its exact Bernstein coefficients are positive; the least exceeds $7/100$. The case $a = 1$ again follows directly from (10). This proves degree ten.

6 Degree eleven

Here $N = 10$. We use

$$m_{11} = \frac{563}{1000} < 2 \sin \frac{\pi}{11}.$$

Put $x = 571/2000$. Machin's identity and the alternating series for arctan give

$$\pi = 16 \arctan \frac{1}{5} - 4 \arctan \frac{1}{239} > 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} \right) - \frac{4}{239} = \frac{281476}{89625} > \frac{6281}{2000} = 11x.$$

Thus $x < \pi/11$. Since $\sin t > t - t^3/6$ for $0 < t < 1$,

$$2 \sin \frac{\pi}{11} > 2 \left(x - \frac{x^3}{6} \right) = \frac{563}{1000} + \frac{5830589}{24000000000} > \frac{563}{1000}.$$

The coefficient majorants are

m	2	3	4	5	6	7	8	9	10
c_m	5	10	1447/50	32	3125/108	2121/100	3125/256	509/100	1.

(27)

Here c_5, c_6, c_8 are the exact Cauchy bounds; c_4, c_7, c_9 are rational upper bounds for (3); and the remaining cases follow from Lemma 2.4. (An earlier version of this paper took $c_4 = 70/3$ from (4); the present certificates use the weaker Cauchy value $c_4 = 1447/50 \geq \sqrt{10^{10}/(4^4 \cdot 6^6)}$, which they verify directly.)

For $0 \leq a \leq 27/100$,

$$\prod_{k=1}^{10} r_k \leq \left(\frac{127}{100} \right)^{10} < 11.$$

On $[27/100, 37/100]$, Lemma 2.2 with $M = 137/100$ has $\nu = 10$ and gives

$$S_0 = 9 \left(\frac{100}{137} \right)^2 + \left(\frac{(137/100)^9}{11} \right)^2.$$

The quadratic $S_0 a^2 - 10a + 10 - S_0$ has positive Bernstein coefficients on this interval, so (10) gives $u > 1$.

For $i = 37, \dots, 89$, define $\alpha_i, \beta_i, S_i, \lambda_i, Y_i$ exactly as in the degree-ten proof, with $N = 10$, $m = m_{11}$, and $C = 11$. Divide the vertical interval $[0, Y_i]$ into ten equal parts. On each of the resulting 530 rectangles, use Proposition 3.2 with one value from the fixed menu

$$\mathcal{T} = \left\{ 1, \frac{19}{20}, \frac{9}{10}, \frac{17}{20}, \frac{4}{5}, \frac{3}{4}, \frac{7}{10}, \frac{13}{20}, \frac{3}{5}, \frac{11}{20}, \frac{1}{2} \right\}.$$

For every rectangle at least one choice of $\tau \in \mathcal{T}$ gives a polynomial with all tensor-product Bernstein coefficients positive. The polynomials have bidegree at most $(11, 10)$. The least coefficient among all 530 certificates is greater than $1/4000$. The value of τ recorded for each rectangle in the ancillary certificates is the one maximizing that rectangle's least Bernstein coefficient, and under that selection the least coefficient among all 530 certificates is in fact greater than $1/250$; the bundled verifier instead accepts the first τ of a fixed preference order that already succeeds, and its minimum, though smaller, still exceeds $1/4000$.

For $9/10 \leq a \leq 1$, take the bound S_b from Lemma 2.2 with $M = 2$ and set

$$\lambda_b = \frac{17}{20}.$$

Localization gives

$$1 - u \leq C_b(1 - a), \quad C_b = \frac{S_b(1 + 9/10) - 10}{10(9/10)},$$

so $\eta^2 \leq 2C_b(1 - a)$. Put $1 - a = x^2$ and $\eta = xY$. The exact inequalities

$$\left(\frac{8}{25}\right)^2 \geq \frac{1}{10}, \quad \left(\frac{17}{10}\right)^2 \geq 2C_b$$

contain every feasible pair in

$$0 \leq x \leq \frac{8}{25}, \quad 0 \leq Y \leq \frac{17}{10}.$$

Use the unsplit polynomial (19); after division by x^2 , its exact Bernstein coefficients are positive on this box;

the least exceeds $3/50$. This proves degree eleven.

7 Exact verification

If

$$R(U, V) = \sum_{k=0}^r \sum_{\ell=0}^s a_{k\ell} U^k V^\ell,$$

then its tensor-product Bernstein coefficients of bidegree (r, s) are

$$b_{ij} = \sum_{k \leq i} \sum_{\ell \leq j} a_{k\ell} \frac{\binom{i}{k}}{\binom{r}{k}} \frac{\binom{j}{\ell}}{\binom{s}{\ell}}. \quad (28)$$

The ancillary verifiers construct every integral in (19) and (24) symbolically, apply the affine map from the relevant rational rectangle to $[0, 1]^2$, use (28), and compare rational numbers exactly. No floating-point arithmetic or numerical quadrature enters any certificate.

For each degree there are two independent implementations. The primary implementation uses Python integers and rational fractions; the second reconstructs the polynomials independently in SymPy. The finite verification is summarized in Table 2.

Table 2: Exact certificate summary.

Degree	Interior region	Rectangles	Interior lower bound	Boundary lower bound
9	$[9/20, 9/10]$	18	$1/2100$	$1/100$
10	$[2/5, 9/10]$	50	$1/3100$	$7/100$
11	$[37/100, 9/10]$	530	$1/4000$	$3/50$

The certificate programs and a verification log are included with the source of this paper.

8 Conclusion

The reciprocal normalization, localization inequality, and integral identity reduce Sendov’s conjecture in degrees nine, ten, and eleven to exact low-dimensional polynomial inequalities. The degree-ten proof is obtained from the degree-nine framework by sharper centered symmetric-function bounds. Degree eleven requires only one additional estimate, namely the split arithmetic–geometric mean bound in Proposition 3.2. Theorem 1.1 follows.

References

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